“SALES AND REVENUE ANALYSIS”

A PROJECT REPORT

Submitted by

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(220760107029)

In partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

In

COMPUTER ENGINEERING

Shree Swami Atmanand Saraswati Institute of Technology, Surat

Gujarat Technological University, Ahmedabad

APRIL 2026

Shree Swami Atmanand Saraswati Institute of Technology

Surat

CERTIFICATE

This is to certify that the project report submitted along with the project entitled **"Sales And Revenue Analysis"** has been carried out by **Donda Avi Sureshbhai (220760107029)** under my guidance in partial fulfillment for the degree of Bachelor of Engineering in Computer Engineering, 8th Semester of Gujarat Technological University, Ahmedabad during the academic year 2025- 26.

Prof. Radha Bhanderi	Prof. Chirag R Patel
Internal Guide	Head of the Department

JOINING LETTER

ANVESHANA AI

8 January 2026

Dear AVI SURESHBHAI DONDA ,

Congratulations! We are delighted to extend an offer to you to join our company AnveshanaAI in the capacity of a Data Science intern. AnveshanaAI has been growing rapidly and we believe your skills and capabilities will be a valuable addition to our team. We are excited about the prospect of you bringing your unique insights and experience to our workspace.

We are thrilled about the opportunity to work with you and are confident that your contributions will play a key role in our growth. We trust that this role will be a rewarding experience for you, providing an opportunity to broaden your knowledge and skills in Data science. Further details of your orientation and additional information will be communicated to you via mail shortly. We are eagerly waiting for you to become a part of our team and contribute to our shared success.

Looking forward to your affirmative response and to welcome you to our dynamic team.

Sincerely,

Shashank Chaudhary
CEO

+91 6353140359
www.anveshanaai.in
shashank.chaudhary@anveshanaai.in
F-106,Sumel Business park 6, Dudheshwar road, Sahibaug, Ahmedabad

INTERNSHIP CERTIFICATE

ANVESHANA AI

INTERNSHIP COMPLETION CERTIFICATE

This is to certify that **Mr. DONDA AVI SURESHBHAI,** student of **Shree Swami Atmanand Saraswati Institute of Technology, Surat** has successfully completed her internship as **Data Science Intern** with **AnveshanaAI** during the periodo **19th January 2026** to **19th April 2026.**

During the course of internship Avi has shown great amount of responsibility, sincerity and a genuine willingness to learn and zeal to take on new assignments & challenges and delivered good results.

We wish him all the very best for his future.

Sincerely,

Shashank Chaudhary
CEO

+91 6353140359

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Shree Swami Atmanand Saraswati Institute of Technology

Surat

DECLARATION

I hereby declare that the Internship report submitted along with the Internship entitled **"Sales And Revenue Analysis"** submitted in partial fulfillment for the degree of Bachelor of Engineering in to Gujarat Technological University, Ahmedabad, is a bonafide record of original project work carried out by me at under the supervision of and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

DONDA AVI SURESHBHAI

(220760107029)

ACKNOWLEDGMENT

I would like to express my sincere gratitude to all those who have contributed, directly or indirectly, to the successful completion of my internship work. This achievement would not have been possible without the guidance, support, and encouragement of many individuals.

I extend my heartfelt appreciation to my university **SHREE SWAMI ATMANAND SARASWATI INSTITUTE OF TECHNOLOGY (SSASIT),** Surat, for providing me with the opportunity and resources to undertake this internship.

I am deeply indebted to **Prof. Radha Bhanderi**, my internship mentor from SSASIT, Surat, for her invaluable guidance, insightful suggestions, and continuous encouragement. Her dedication to excellence and willingness to share knowledge greatly inspired me and played a crucial role in shaping my work.

I would also like to express my sincere thanks to **Mr. Shashank Choudhary**, Team Leader and my Industrial Mentor, for his constant support, guidance, and motivation throughout the internship. His practical insights and encouragement significantly enhanced my learning experience.

I am grateful to **Dr. Jignesh G. Vaghasiya,** Principal of SSASIT, Surat, and **Prof. Chirag Patel**, Head of the Department of Computer Engineering, along with all the faculty members and staff, for their support and cooperation.

I would also like to take this opportunity to thank my parents, guardians, and friends for their unwavering support, encouragement, and understanding throughout this journey.

Finally, I express my sincere thanks to all known and unknown individuals who have, in one way or another, contributed to the completion of this work.

DONDA AVI SURESHBHAI

(220760107029)

ABSTRACT

This project focuses on analyzing real-world data to extract meaningful insights and support data-driven decision-making. It covers the complete data analysis process, starting with data collection from reliable sources, followed by data cleaning and preprocessing to ensure accuracy and consistency. Key steps included handling missing values, removing duplicates, and organizing the data into a structured format suitable for analysis. Exploratory Data Analysis (EDA) was performed using Python libraries such as NumPy and Pandas for data manipulation, along with Matplotlib for visualization. Through EDA, patterns, trends, and relationships within the dataset were identified, helping to better understand the data and uncover important insights. Basic statistical techniques were also applied to summarize the data and highlight key observations. SQL was used to query structured data efficiently, enabling filtering, aggregation, and extraction of relevant information for analysis. The findings were presented using clear visualizations and dashboards, making the results easy to interpret for both technical and non-technical audiences. Overall, the project highlights practical skills in data analysis, including data cleaning, visualization, and querying, while also emphasizing business understanding and effective communication of insights.

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1. . COMPANY OVERVIEW

1.1 ABOUT ANVESHANAAI

- It is a Prominent IT firm with 5 years of experience and knowledge. It has a group of over 15 developers with years of experience in the field
- Company helps other companies from small scale to MNCs to make their dream come true by transforming the client's ideas and achieving new milestones with their digital and social media marketing services
- Builds AI-powered solutions for academic evaluation, research analysis, and institutional workflow automation.

1.2 COMPANY VISION

- Empowering businesses & individuals to achieve growth and prosperity in the Digital realm.

1.3 COMPANY MISSION

- Expand client's business reach within the industry by delivering cutting-edge using ai that meet their needs with complete satisfaction and transparency.

2. OTHER DEPARTMENTS OF COMPANY

2.1 Research & Development (R&D) Department

- This department focuses on developing and improving AI models used for academic evaluation, plagiarism detection, and research gap analysis. It continuously works on innovation to enhance the platform's accuracy and efficiency.

2.2 Product Development

- The product development team designs and builds the core platform, including dashboards, tools, and system features. They ensure the platform is user-friendly, scalable, and meets the needs of educational institutions.

2.3 Quality Assurance (QA) & Testing

- This department is responsible for testing the software to ensure it is reliable, secure, and error-free. It validates system performance and ensures that all features work smoothly before deployment.

2.4 Sales & Marketing

- The sales and marketing team promotes the platform to universities and institutions. It focuses on client acquisition, brand awareness, and communicating the benefits of AI-driven academic solutions.

3. INTERNSHIP MANAGEMENT

3.1 INTERNSHIP MANAGEMENT

- Gained practical exposure to working in an AI-based company environment, including understanding system workflows, data handling processes, and dashboard creation.

3.2 PURPOSE

- The main purpose of the internship was to gain real-world industry experience, understand how real world data analysis done, and improve professional and technical skills for future career growth.

3.3 OBJECTIVE

- To understand the fundamentals of data analysis and develop practical skills using Microsoft Excel for handling, processing, and analyzing datasets.
- To apply data cleaning, transformation, and visualization techniques to derive meaningful insights from raw data.
- To enhance analytical thinking and problem-solving skills through real-world data analysis tasks.

3.4 SCOPE

- The scope of this internship was to understand the complete data analysis process, including data collection, cleaning, and visualization using Excel.
- To study how organizations use data for decision-making and performance evaluation.
- To contribute to improving data organization, reporting efficiency, and insight generation through structured analysis technique

3.5 TECHNOLOGY USED

- **Technologie/Concepts:** Python, Excel,Power bI .
- **Tools:** Jupyter notebook, Microsoft excel, Microsoft Power BI

3.6 INTERNSHIP DEVELOPMENT APPROACH

- In the initial phase of the internship, I focused on understanding the company's platform, workflow, and tools used in the AI-based environment. I observed how different components of the system interact and how processes are managed.
- In the later phase, I analyzed system functionality, identified areas of improvement, and enhanced my understanding of real-world implementation practices in a professional environment.

3.7 INTERNSHIP EFFORT AND TIME, COST ESTIMATION

- My Whole effort was behind analyzing data and learning tools like excel ,power bi.
- Time of internship: 5 days / week (6.00 working hours)
- Total time of internship (In weeks): 13 week

4. PYTHON FOR DATA ANALYSIS

4.1 Introduction

- Python is a high-level programming language widely used for data analysis due to its simplicity and powerful libraries. It allows users to process, analyze, and visualize large datasets efficiently.
- It provides powerful libraries like Pandas, NumPy, and Matplotlib for handling and analyzing data..
- Python is easy to learn and supports readable and structured coding practices.
- It is widely used in industries for data processing, automation, and AI-based applications.
- Python allows seamless integration with databases, APIs, and visualization tools, making it highly versatile.
- It supports rapid development and prototyping, which helps developers build applications quickly and efficiently.

4.2 Features

- Simple and easy-to-understand syntax
- Large number of data analysis libraries
- Supports data visualization
- Platform independent
- Strong community support
- Integration with AI and machine learning tools

4.3 Advantages

- Python has simple syntax compared to other programming languages. This makes it beginner-friendly and easy to understand. It reduces development time.
- Python provides libraries like Pandas and NumPy for data manipulation. These tools simplify complex data operations. They improve efficiency in analysis.
- Libraries like Matplotlib and Seaborn help in creating graphs and charts. Visualization makes data easier to understand. It helps in better decision-making.
- Python can be used for multiple purposes like web development, AI, and data analysis. This makes it a versatile language. It adapts to various project need.
- Python has a large developer community. Many tutorials and resources are available online. This helps in solving problems quickly.

4.4 Disadvantages

- Python is slower compared to compiled languages like C++. This can affect performance in large applications. It may not be suitable for high-speed systems.
- Python consumes more memory compared to other languages. This can be an issue in memory-sensitive applications. Optimization is sometimes required.
- Python is not commonly used for mobile app development. Other languages like Java or Swift are preferred. This limits its usage in mobile platforms.
- Python is dynamically typed, which can lead to runtime errors. Errors may occur during execution. Proper testing is required.
- Python has limitations in multi-threading due to GIL. This affects performance in parallel processing. It is not ideal for CPU-heavy tasks.

4.5 Minimum requirements

- System with Windows/Linux/Mac OS
- Minimum 4 GB RAM (8 GB recommended)
- Python installed (latest version)
- Code editor like VS Code or Jupyter Notebook
- Basic knowledge of programming

5 . SETUP AND CREATION

- Download and install Python from official website

  https://nodejs.org/en/download/

- Install Code Editor VS Code or Jupyter Notebook

  https://reactnative.dev/docs/environment-setup

- Install Required Libraries

Pandas

  - Used for data manipulation and analysis
  - Works with tables (DataFrames)
  - pip install pandas

NumPy

  - Used for numerical calculations and arrays
  - pip install pandas numpy matplotlib

Matplotlib

  - Used for data manipulation and analysis
  - Works with tables (DataFrames)
  - pip install matplotlib

- Create Python File / Notebook
  - Use .py file in VS Code OR
  - .ipynb file in Jupyter Notebook

- Load Dataset
  - Import CSV or Excel file
  - Example:

```
import pandas as pd
data = pd.read_csv("file.csv")
```

- Data Cleaning & Preprocessing
  - Remove missing values
  - Convert data types
  - Filter unwanted data

- Analyze data using functions and libraries
- Visualize data using charts and graphs
- Save results and conclusions

6. BASIC COMPONENTS

- **Pandas**
  - ➢ Used for data manipulation and analysis. It provides DataFrame structure for handling data.

- **NumPy**
  - ➢ Supports numerical operations and array processing. It improves performance in calculations.

- **Matplotlib**
  - ➢ Used for creating basic graphs and charts. Helps in visual representation of dat.

- **Seaborn**
  - ➢ Advanced visualization library built on Matplotlib. Used for statistical graphs.

- **Jupyter Notebook**
  - ➢ Interactive environment for writing and running Python code. Useful for analysis and documentation.

- **DataFrame**
  - ➢ A table-like structure in Pandas. Used to store and manipulate data efficiently.

- **CSV/Excel Files**
  - ➢ Common data sources used in analysis. Easily loaded into Python.

- **Functions**
  - ➢ Reusable blocks of code. Help in organizing and simplifying analysis tasks.

7. EXCEL FOR DATA ANALYSIS

7.1 What is Excel?

- Microsoft Excel is a widely used spreadsheet application developed by Microsoft. It is used for data storage, organization, analysis, and visualization. Excel provides powerful tools such as formulas, functions, charts, and pivot tables, making it suitable for handling both simple and complex datasets in business environments.

7.2 Data Collection and Importing

- Excel allows users to collect and import data from various sources, including:
  - Manual data entry
  - CSV (Comma Separated Values) files
  - Databases and external sources
  - Online data sources
- Data can be imported using features like "Get & Transform" (Power Query), which helps in efficiently loading large datasets into Excel.

7.3. Data Cleaning and Transformation

- Data cleaning is an essential step before analysis. In Excel, this includes:
  - Removing duplicates
  - Handling missing values
  - Correcting formatting errors
  - Using Text-to-Columns
- Transformation techniques include sorting, filtering, and applying formulas to restructure the dataset for analysis.

7.4 Data Modeling and Visualization

- Excel supports data modeling through:
  - Use of formulas (IF, VLOOKUP, INDEX-MATCH)
  - Pivot Tables for summarizing data
  - Data relationships (in advanced Excel)
- Visualization tools include:
  - Bar charts, line charts, pie charts
  - Conditional formatting
  - Sparklines
- These tools help in understanding patterns and trends in the data.

7.5 Dashboard Creation and Insights

- Excel enables the creation of interactive dashboards using:
  - Pivot Charts
  - Slicers and Timelines
  - Dynamic charts
- Dashboards provide insights such as:
  - Performance tracking
  - Trend analysis
  - Key business metrics (KPIs)

7.6 Advantages of Excel

- Easy to use and widely available
- Supports a wide range of functions and formulas
- Suitable for small to medium-sized datasets
- Quick visualization and reporting
- Integration with other tools like Power BI

8 . Power BI For Data Analysis

8.1 What is Power BI?

- Power BI is a powerful business intelligence and data visualization tool developed by Microsoft.
- It is used to convert raw data into meaningful insights through interactive dashboards and reports.
- Power BI allows users to connect to multiple data sources such as Excel, databases, and cloud services.
- It helps in analyzing data and presenting it in a visually appealing and easy-to-understand format.

8.2 Data Collection and Importing

- Data was collected from various sources such as Excel files, CSV files, and databases.
- Power BI provides options to easily import and connect data from multiple platforms.
- This step ensures that all required data is available for analysis in a single environment.

8.3 Data Cleaning and Transformation

- Data cleaning was performed using Power Query Editor.
- Tasks included removing null values, correcting data types, filtering unnecessary data, and reshaping datasets.
- Proper data transformation ensures accuracy and consistency in analysis.

8.4 Data Modeling and Visualization

- Relationships were created between multiple tables to build a structured data model.
- DAX (Data Analysis Expressions) was used to create calculated columns and measures.
- Various visualizations such as bar charts, line charts, pie charts, and tables were used.

- Slicers and filters were added to make dashboards interactive and user-friendly.

8.5 Dashboard Creation and Insights

- Interactive dashboards were created to present data insights clearly.
- A sales dashboard was developed to analyze trends, top-performing products, and regional performance.
- Key Performance Indicators (KPIs) like total sales, profit, and growth rate were included.
- Power BI supports report sharing and real-time updates, improving collaboration.
- Overall, Power BI enhanced my ability to present data effectively and strengthened my visualization skills.

8.6 Advantages of Power BI

- Power BI provides an easy-to-use interface for creating interactive dashboards and reports.
- It supports integration with multiple data sources such as Excel, SQL databases, and cloud services.
- It enables real-time data updates, allowing users to access the latest information.
- The use of DAX functions allows advanced data analysis and calculations.
- Power BI dashboards are highly interactive with features like slicers and filters.
- It helps in better decision-making by presenting data in a clear and visual format.
- Reports can be easily shared with team members, improving collaboration and efficiency.

9 . INTERNSHIP ACTIVITIES

WEEK	DATE	ACTIVITY
Week 1	19/01 – 25/01	Introduction to data analytics
Week 2	26/01 – 01/02	Learning python
Week 3	02/02 – 08/02	Learning different libraries of python
Week 4	08/02 – 15/02	Data visualization using matplotlib
Week 5	16/02 – 22/02	Exploratory data analysis
Week 6	23/02 – 01/03	Learning fundamentals of SQL
Week 7	02/03 – 08/03	Learning probabilities
Week 8	09/03 – 15/03	Learn basic different machine learning concept
Week 9	16/03 – 22/03	Implement different ML models
Week 10	23/03 – 29/03	Learning excel
Week 11	30/03 – 05/04	Learning power BI tool
Week 12	06/04 – 12/04	Start data analysis for sales and revenue performance using python
Week 13	13/04 – 19/04	Made dashboard using excel and power BI

Table 9.1 Week wise Internship activities

10 . IMPLEMENTING PROJECT

```
[2]
✓ 2s

# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("train.csv")

# Display basic info
print("First 5 rows:")
print(df.head())

print("\nDataset Info:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())
```

Fig 10.1 import different libraries

```
Missing Values:
Row ID           0
Order ID         0
Order Date       0
Ship Date        0
Ship Mode        0
Customer ID      0
Customer Name    0
Segment          0
Country          0
City             0
State            0
Postal Code     11
Region           0
Product ID       0
Category         0
Sub-Category     0
Product Name     0
Sales            0
dtype: int64
```

Fig 10.2 handle missing values

[9]

```python
# ----------------------------
#  Data Cleaning
# ----------------------------

# Convert dates to datetime
df['Order Date'] = pd.to_datetime(df['Order Date'], format='%d/%m/%Y')
df['Ship Date'] = pd.to_datetime(df['Ship Date'], format='%d/%m/%Y')


# Create new columns
df['Year'] = df['Order Date'].dt.year
df['Month'] = df['Order Date'].dt.month
df['Month Name'] = df['Order Date'].dt.month_name()
```

Fig 10.3 Data cleaning

```
Total Revenue: 2261536.7827000003
```

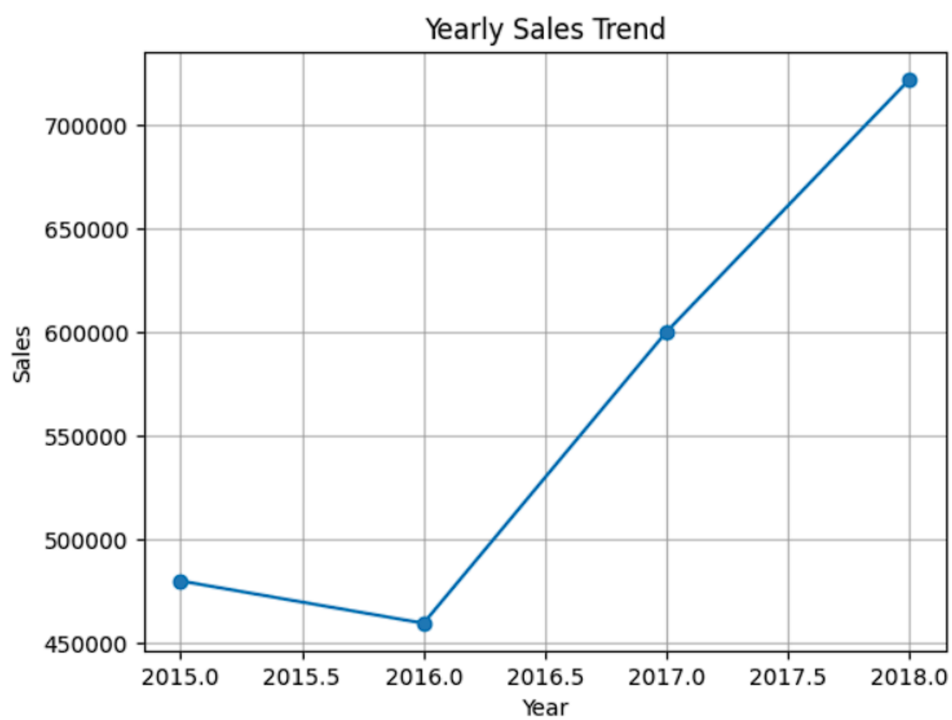

Fig 10.4 Yearly sales trend

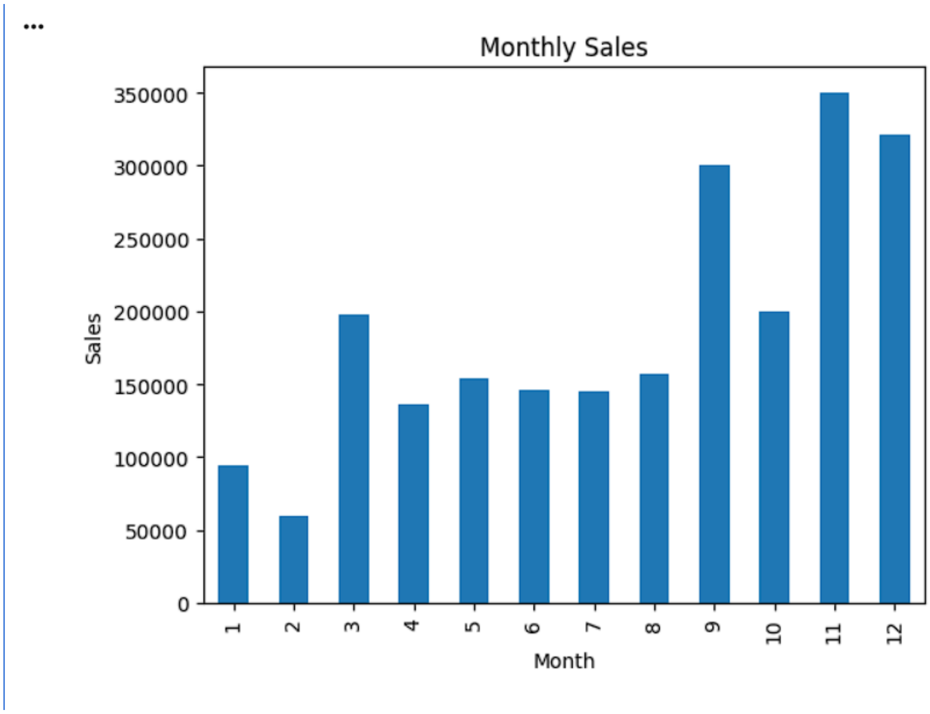

Fig 10.5 Monthly sales

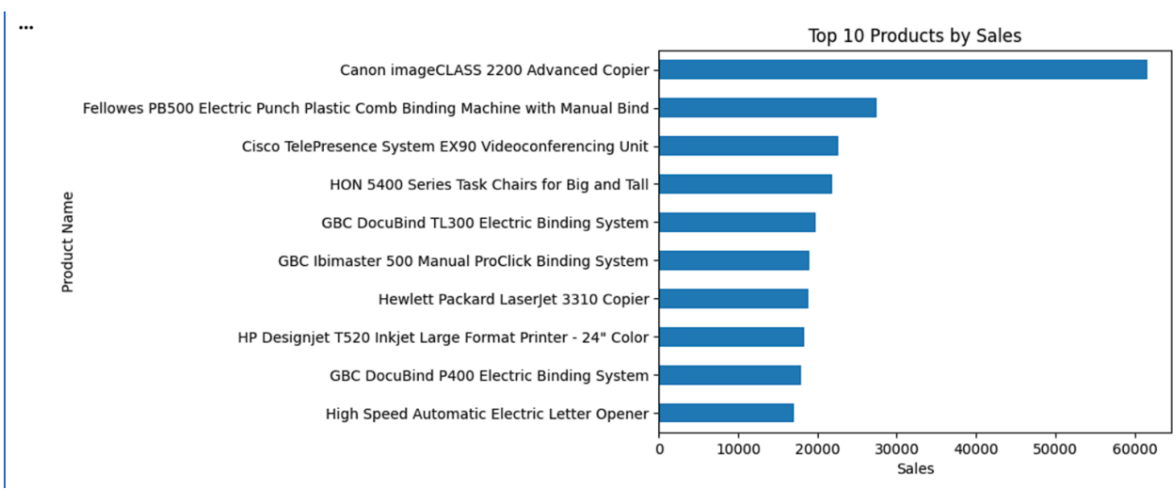

Fig 10.6 Top 10 products by sale

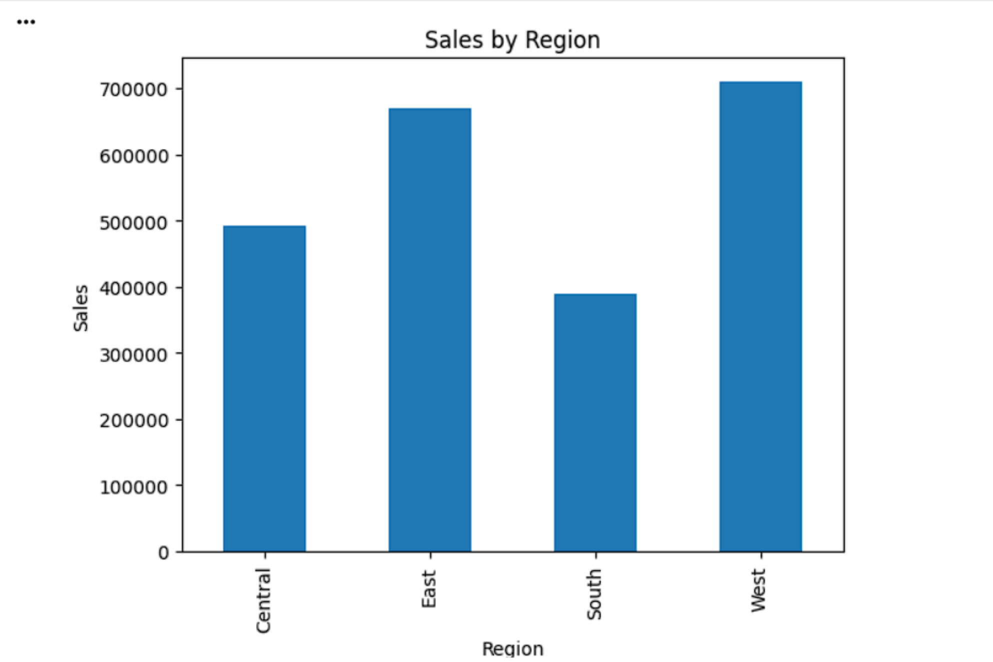

Fig 10.7 Sales by region

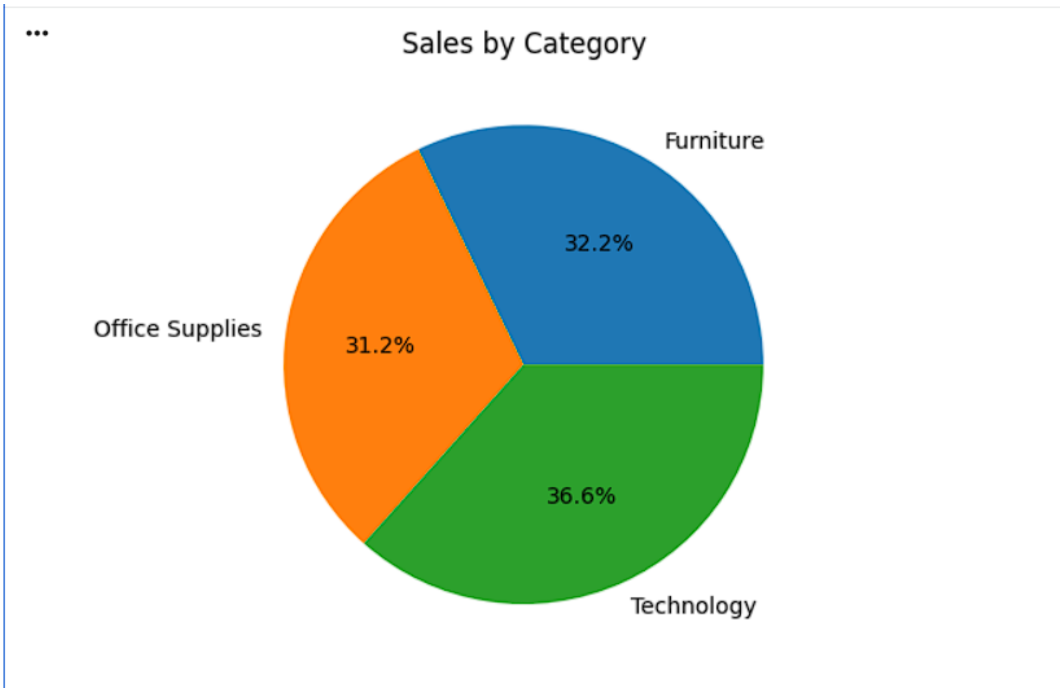

Fig 10.8 Sales by category

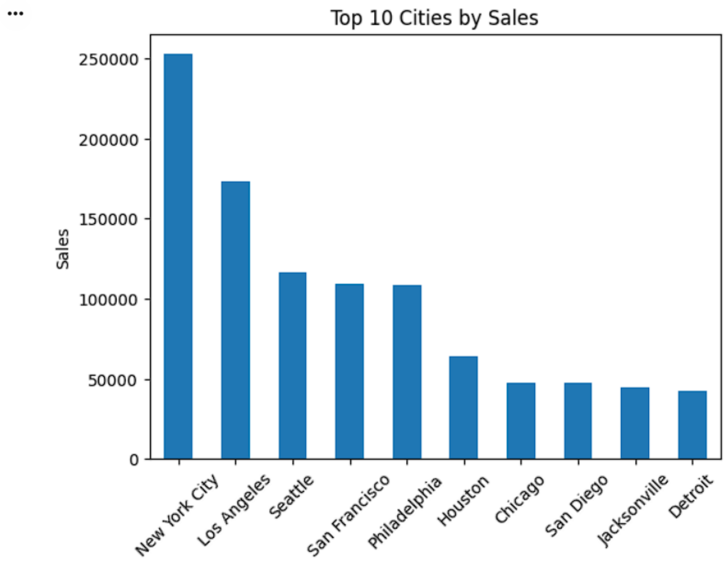

Fig 10.9 Top 10 cities by sales

```python
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, r2_score

# Select better features
features = ['Category', 'Sub-Category', 'Region', 'Segment', 'Ship Mode']
X = df[features]
y = df['Sales']

# One-hot encoding (better than LabelEncoder)
X = pd.get_dummies(X, drop_first=True)

# Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Model
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluate
print("MAE:", mean_absolute_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
```

```
... MAE: 245.95192611955818
R2 Score: 0.10526366627333117
```

Fig 10.10 machine Learning

```
!pip install plotly
Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages (5.24.1)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.12/dist-packages (from plotly) (9.1.4)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from plotly) (26.0)
```

Fig 10.11 install plotly for interactive dashboard

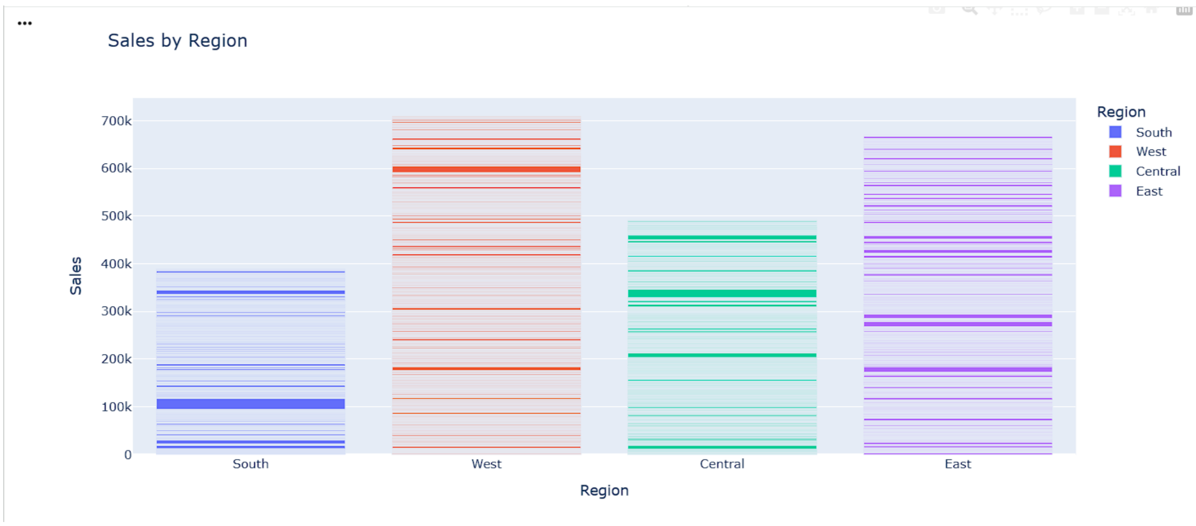

Fig 10.12 Sales by region

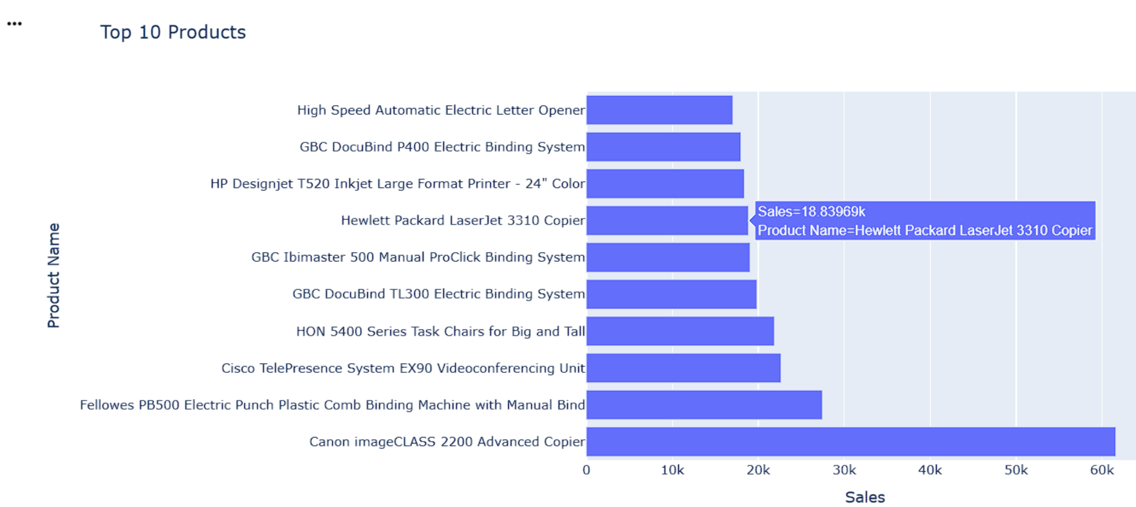

Fig 10.13 Top 10 product

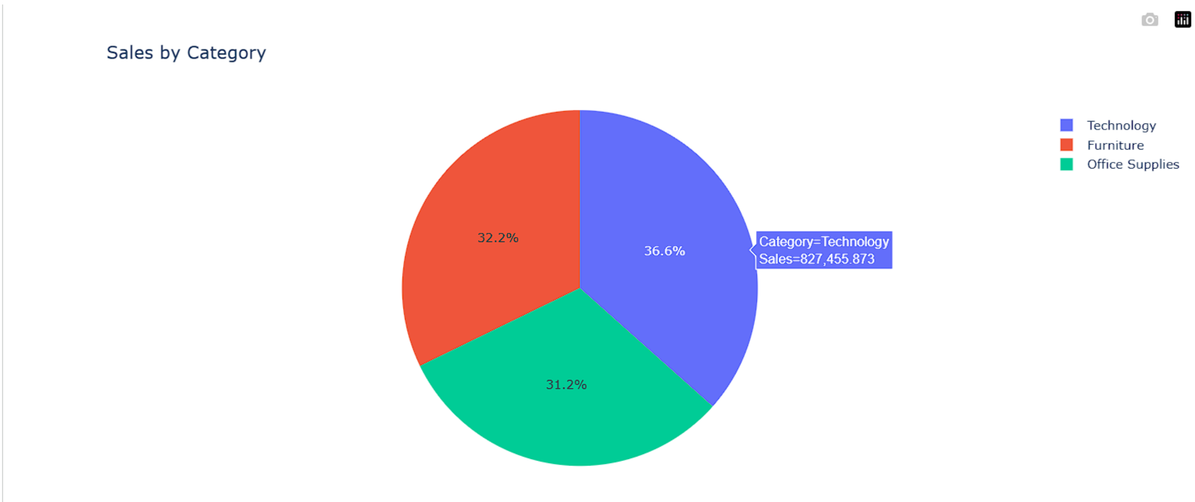

Fig 10.14 Sales by category

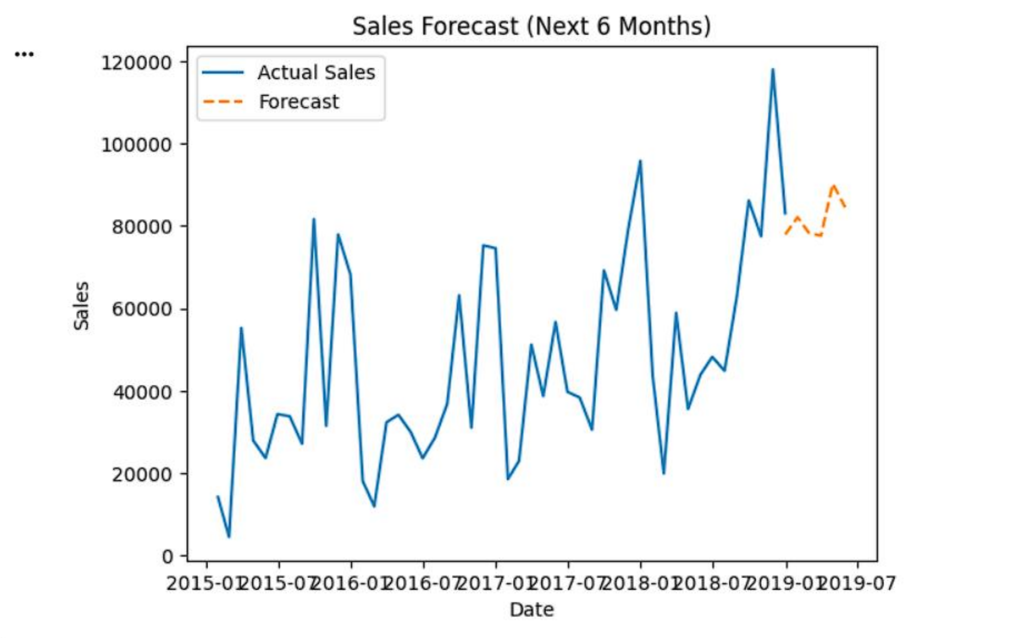

Fig 10.15 Sales forecast

CONCLUSION

In conclusion, the development of this data analysis project has been a valuable learning experience in working with real-world datasets and applying data-driven techniques. The project successfully demonstrates the end-to-end data analysis process, including data collection, cleaning, preprocessing, and exploratory analysis. Tools and technologies such as Python (NumPy, Pandas, and Matplotlib) and SQL were effectively utilized to manage, analyze, and visualize the data.

Through this project, key concepts such as data manipulation, statistical analysis, and visualization were practically implemented to identify patterns, trends, and meaningful insights. The use of SQL further enhanced the ability to efficiently query and extract relevant information from structured datasets.

The final outcomes were presented through clear visualizations and dashboards, ensuring that the insights are easily understandable for both technical and non-technical audiences. Overall, this project strengthened my analytical and technical skills while also improving my ability to communicate insights effectively, providing a strong foundation for future growth in the field of data analysis.

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